

Hybrid physics-ML models for automotive active noise control

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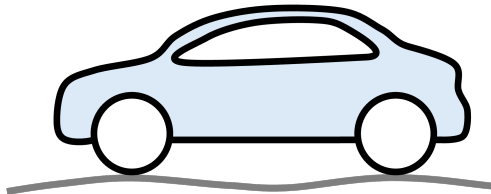


Figure 1: Representation of the major noise pathways in a vehicle.

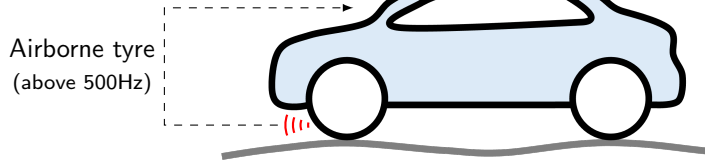


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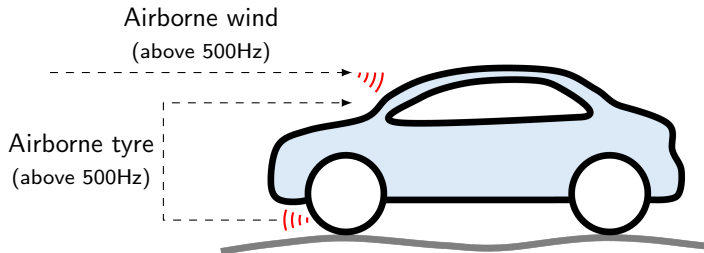


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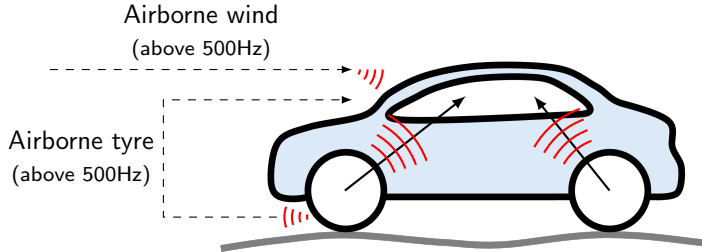


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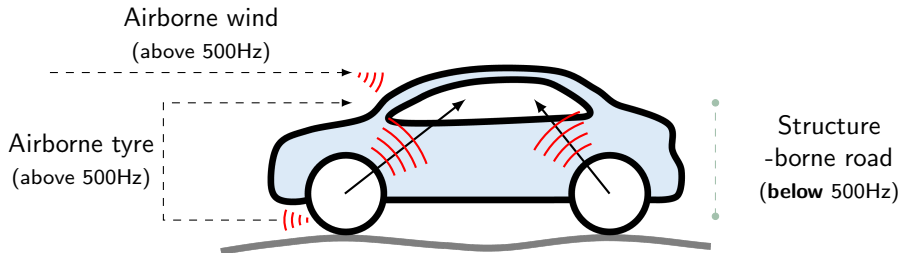


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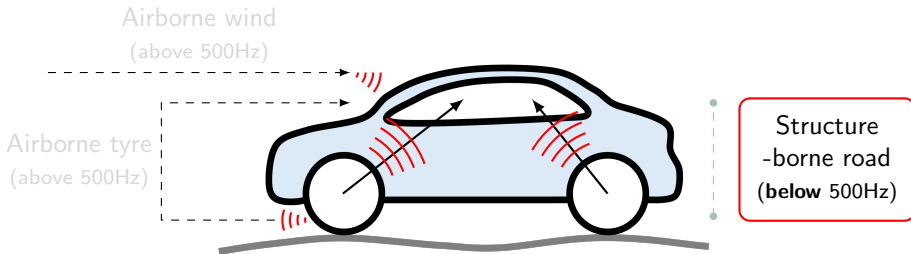


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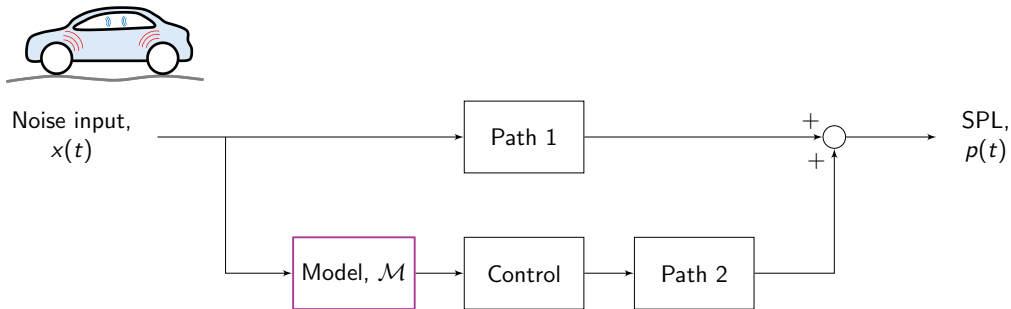


Figure 2: A simplified representation of a feedforward ANC system.

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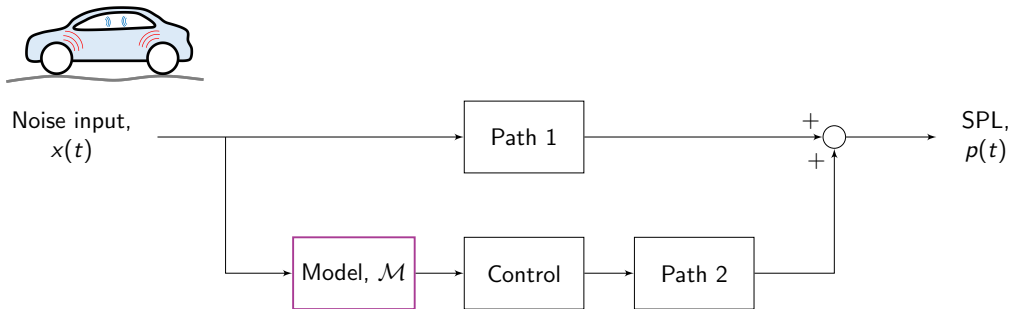


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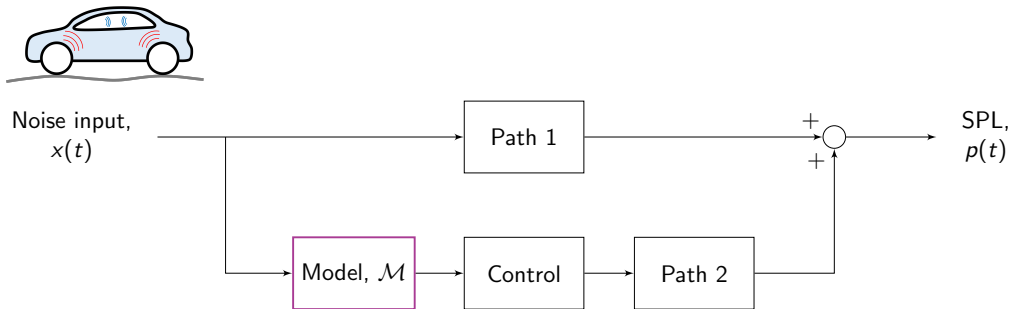


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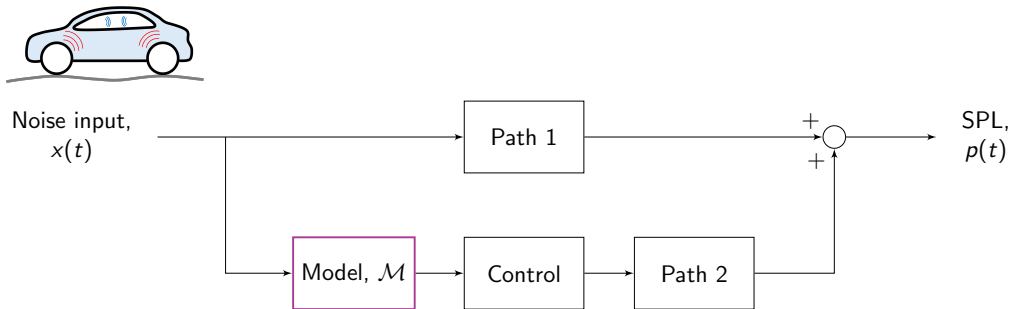
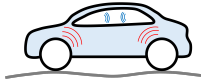


Figure 2: A simplified representation of a feedforward ANC system.

- Therefore, ANC performance is highly dependent on the reference signal model $\mathcal{M}$:
 - ① It must be **fast** - to 'beat' the sound.
 - ② It must be **accurate** - performance is logarithmic.

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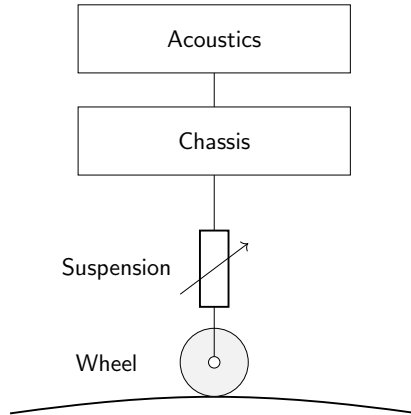
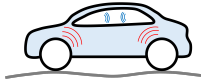


Figure 3: A simplified quarter-car model of the structure-borne noise pathway.

Structure-borne noise

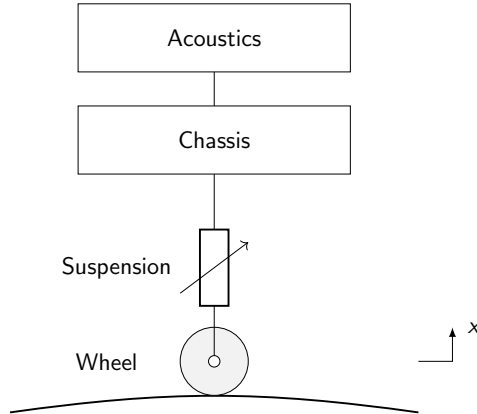
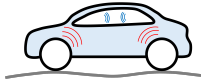


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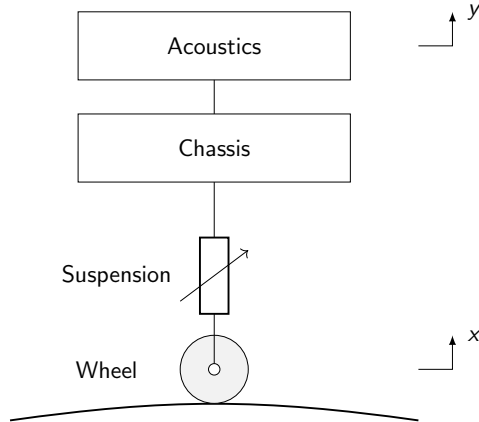
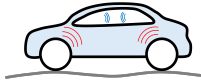
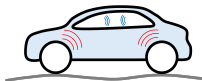


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Linear

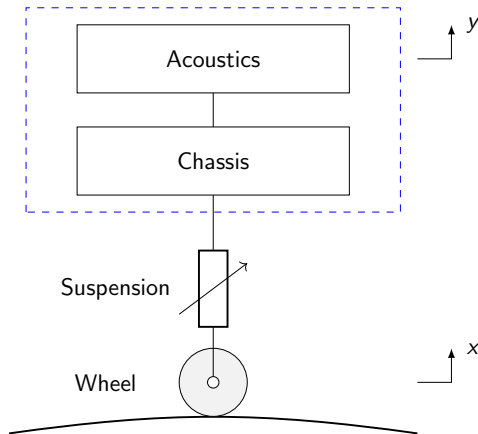


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Linear

Nonlinear

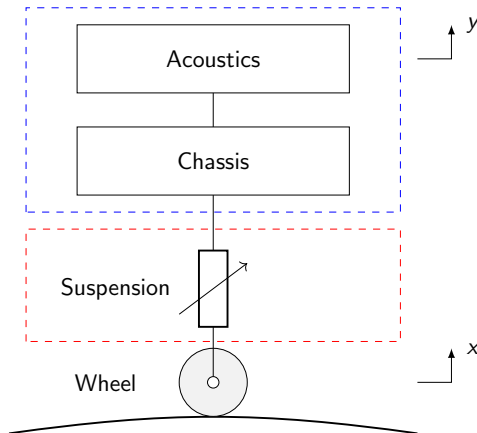


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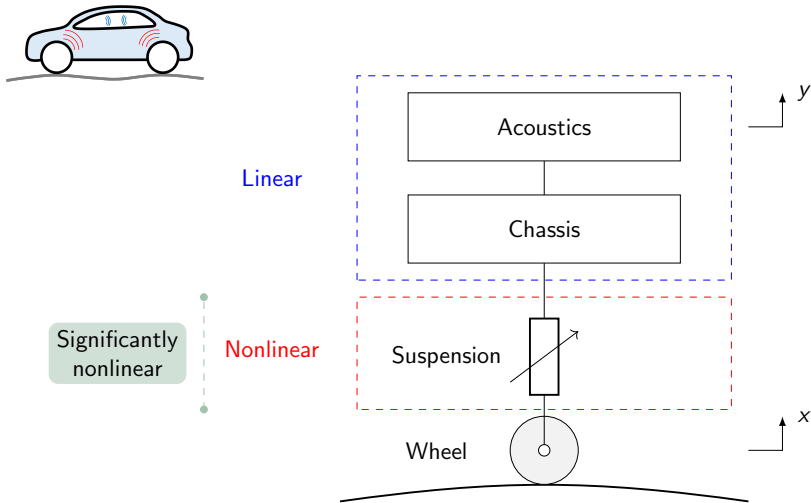


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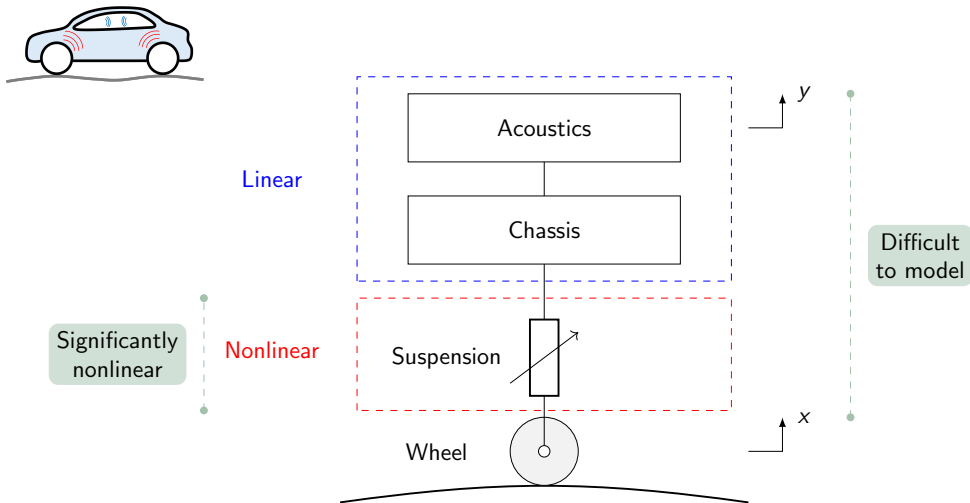


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- 1 Can a hybrid physics-ML model **improve prediction**?

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 - Does including physical knowledge help?

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Hybrid physics-ML models for automotive active noise control

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- Does including physical knowledge help?
- How should we include it?

Hybrid physics-ML models for automotive active noise control

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- Can they be *fast* and *accurate* enough?

Hybrid physics-ML models for automotive active noise control

1 Can a hybrid physics-ML model **improve prediction**?

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2 Are hybrid physics-ML models **suitable for ANC**?

- Can they be *fast* and *accurate* enough?
- How do they scale to real complex systems?

What makes the problem hard?

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¹ M. De Brett, 'Implications of nonlinear suspension behaviour for feedforward control of road noise in cars', PhD Thesis, University of Cambridge, 2021

² J. Massingham, 'Performance Limits of Active Noise Control for Nonlinear Systems: A Machine Learning Perspective', PhD Thesis, University of Cambridge, 2025

What makes the problem hard?

- **Nonlinearity**

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- **Nonlinearity**
 - Causes significant coherence reduction¹ – reduces performance.

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- **Coupling**

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- **Nonlinearity**

- Causes significant coherence reduction¹ – reduces performance.
- We cannot ‘measure it out’ – suspension has multiple connections into the chassis.

- **Coupling**

- Input force is dependent on resulting output displacement.
- Results in a longer system memory – requires more data to learn².

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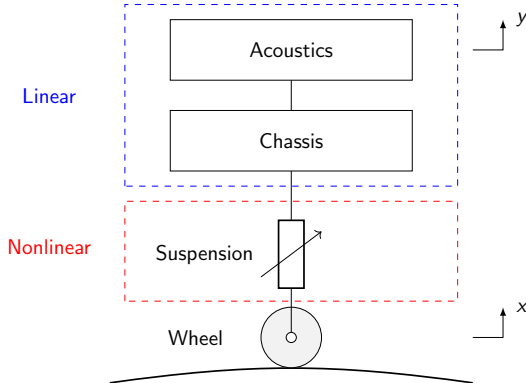


Figure 4: Simplified car.

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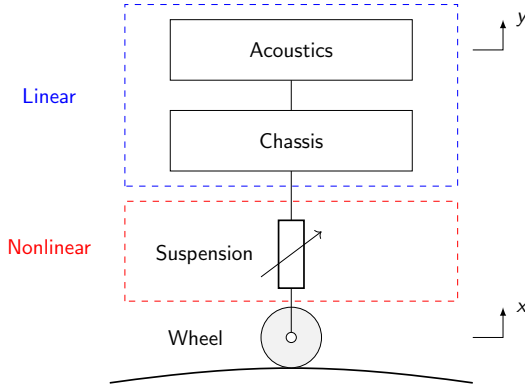


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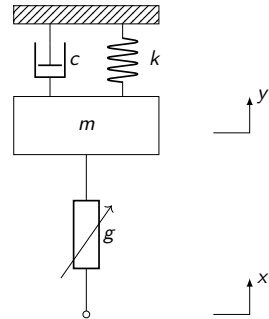


Figure 5: SDOF benchmark system.

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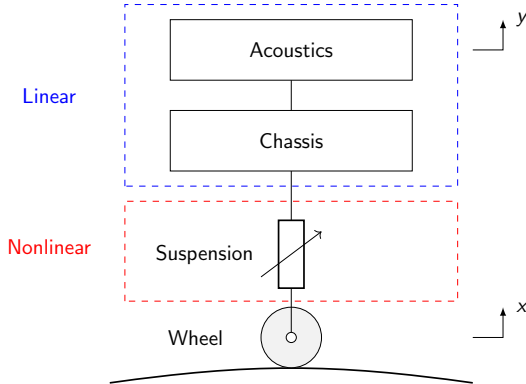


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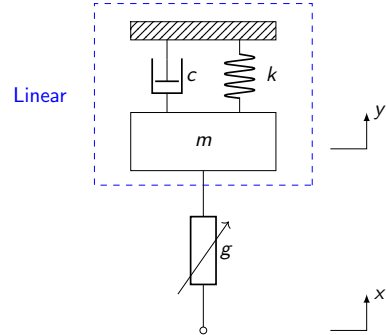


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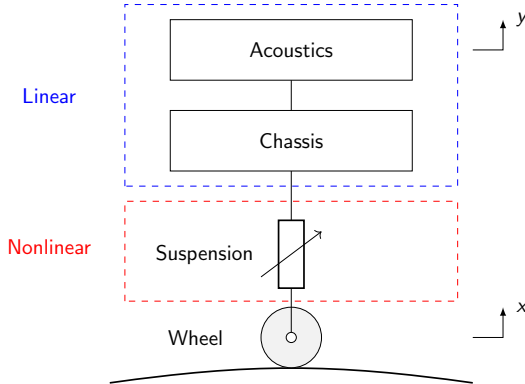


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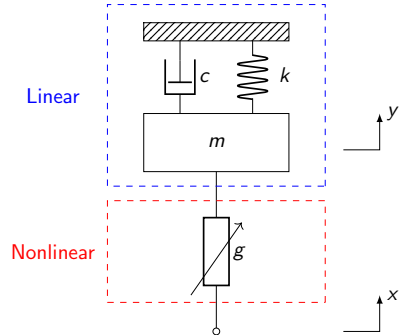


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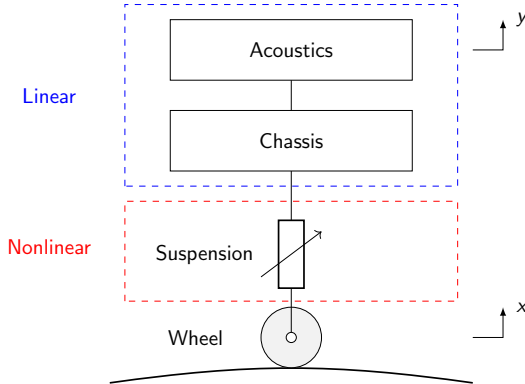


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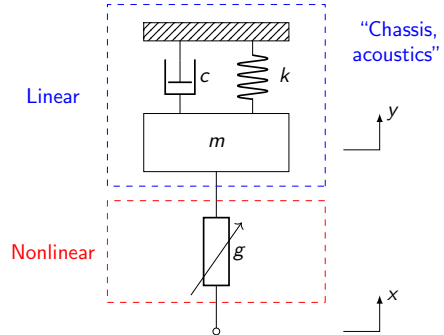


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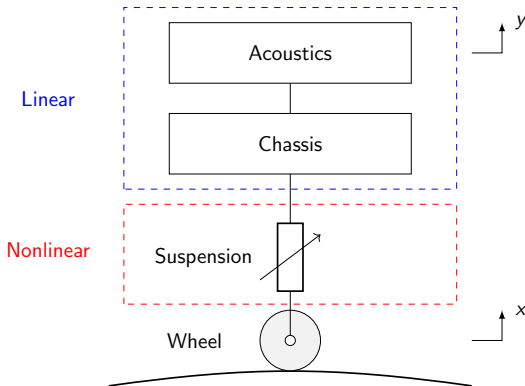


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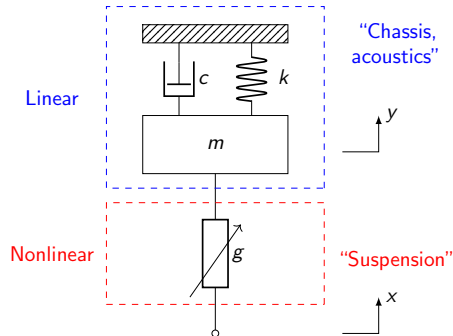


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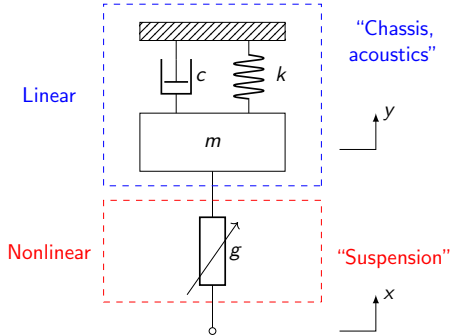


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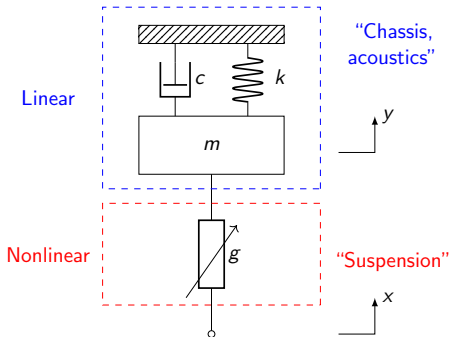
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- SDOF equation of motion:

$$m\ddot{y} + c\dot{y} + ky = g(x - y) \quad (1)$$

Figure 5: SDOF benchmark system.

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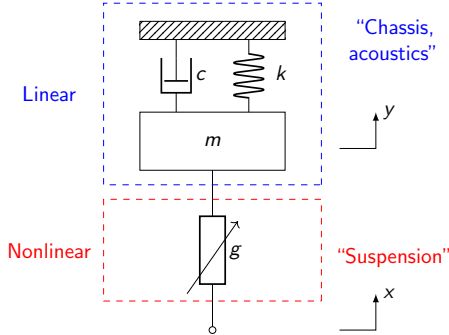
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- SDOF equation of motion:

$$m\ddot{y} + c\dot{y} + ky = g(x - y) \quad (1)$$

MDOF $\Rightarrow$ **M**, **C**, and **K** matrices.

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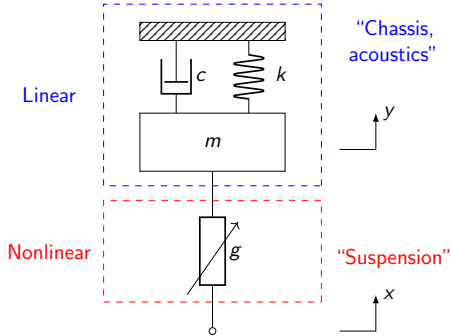
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- SDOF equation of motion:

$$m\ddot{y} + c\dot{y} + ky = g(x - y) \quad (1)$$

MDOF $\Rightarrow$ **M**, **C**, and **K** matrices.

- Rearranging for y :

$$y = \textcolor{blue}{d} * \textcolor{red}{g}(x - y) \quad (2)$$

Figure 5: SDOF benchmark system.

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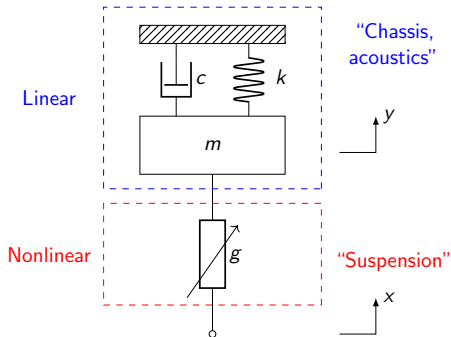


Figure 5: SDOF benchmark system.

- SDOF equation of motion:

$$m\ddot{y} + c\dot{y} + ky = g(x - y) \quad (1)$$

MDOF $\Rightarrow$ **M**, **C**, and **K** matrices.

- Rearranging for y :

$$y = d * g(x - y) \quad (2)$$

d : linear impulse response

g : nonlinearity

$*$: convolution operator

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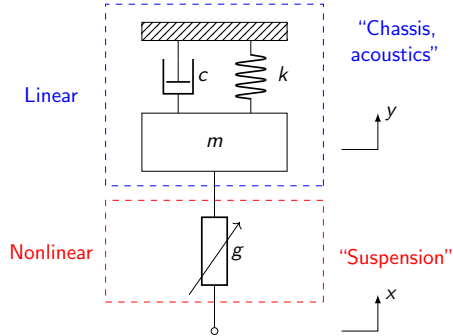


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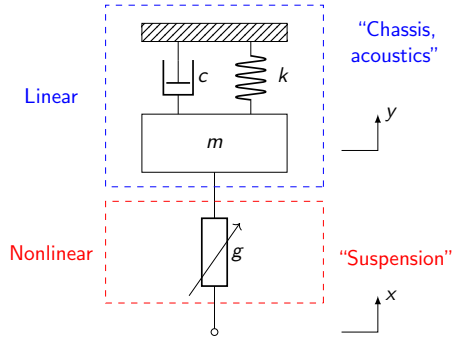
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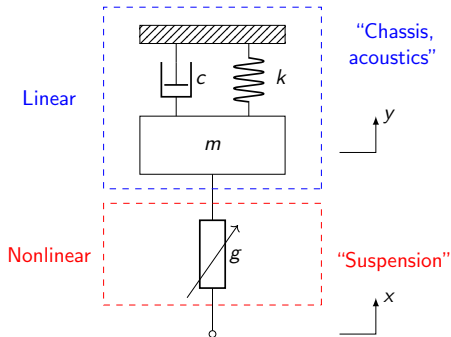


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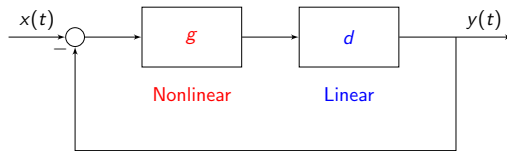


Figure 6: Feedback representation.

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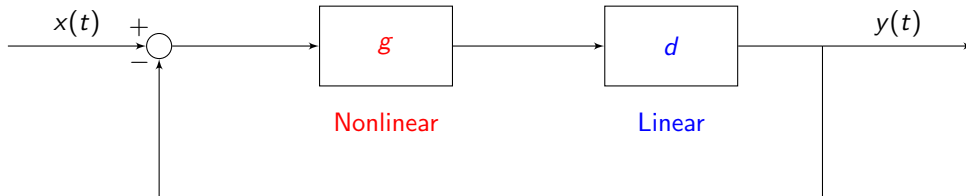


Figure 7: Block diagram of a coupled nonlinear system.

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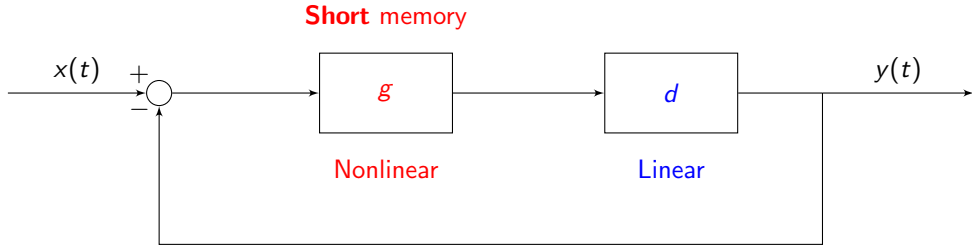


Figure 7: Block diagram of a coupled nonlinear system.

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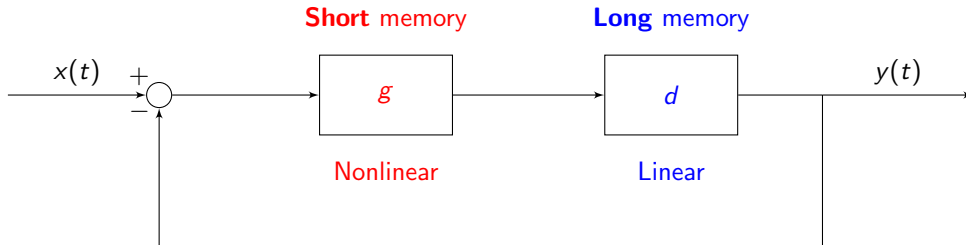


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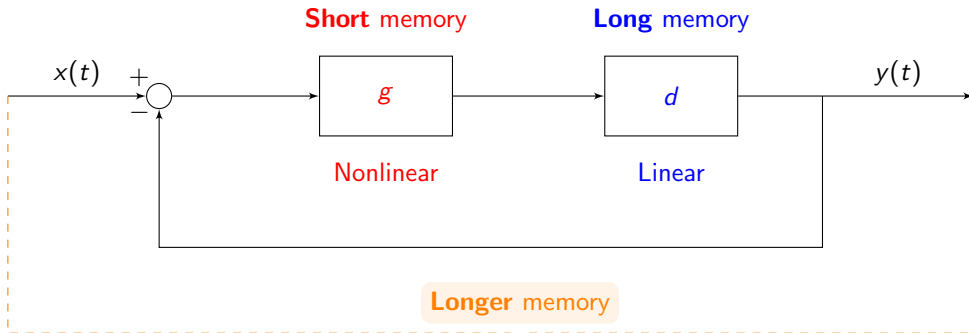


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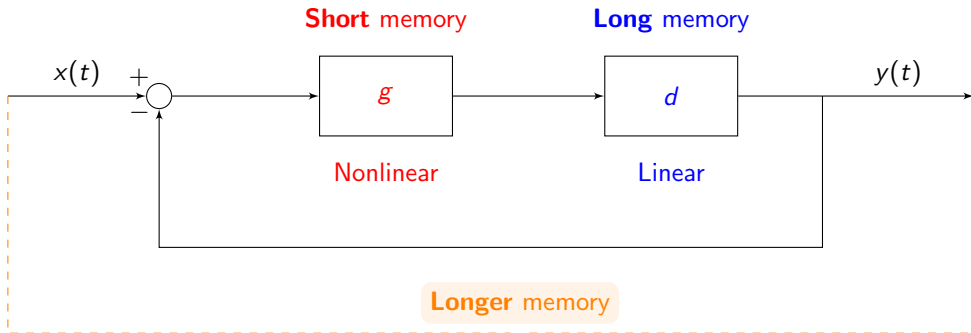


Figure 7: Block diagram of a coupled nonlinear system.

⇒ What if we bake the **feedback structure** into the **model architecture**?

Architecture



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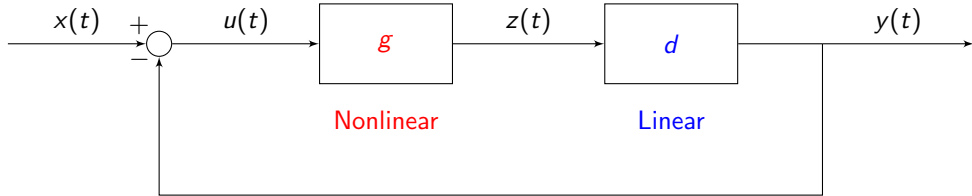


Figure 8: Block diagram of the feedback model architecture.

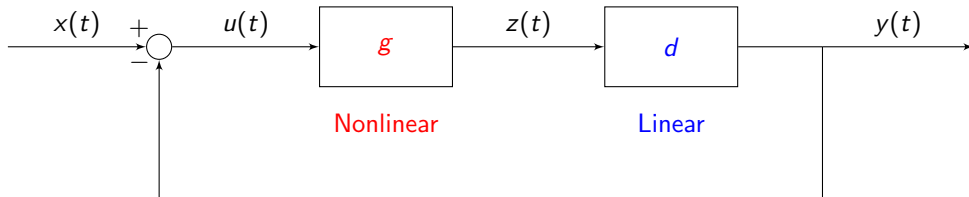


Figure 8: Block diagram of the feedback model architecture.

$$\begin{aligned} u(t) &= x(t) - y(t) \\ y(t) &= d(t) * g(u(t)) \end{aligned} \tag{3}$$

Model components

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- Wide range of model types possible – depending on the choice of d and g .

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Linear model d

- Discrete-time FIR of the MDOF linear system, truncated to L_d samples.

$$d[k] = \sum_{n=0}^D A e^{-\zeta \omega_n k T} \sin(\omega_d k T + \phi) \quad k = 0, 1, \dots, L_d - 1 \quad (4)$$

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- $L_d + L_g \leq$ total memory.

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Comparison: feedforward model

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ANC performance: output coherence

$$\gamma_{y_t y_p}^2(\omega) = \frac{|S_{y_t y_p}(\omega)|^2}{S_{y_t y_t}(\omega) S_{y_p y_p}(\omega)} \quad (6)$$

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- How much of the **nonlinearity is captured?** $\implies$ **achievable noise reduction.**

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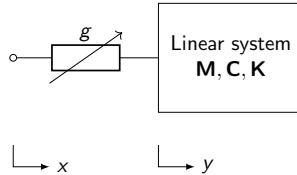


Figure 9: MDOF system used for the benchmark simulations.

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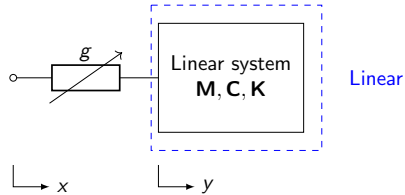


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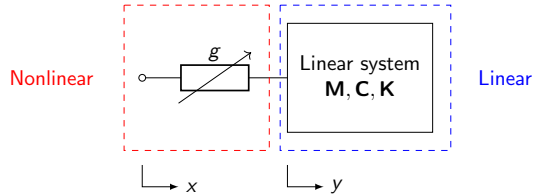


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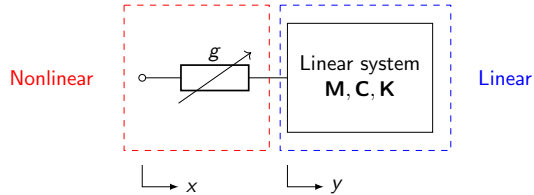


Figure 9: MDOF system used for the benchmark simulations.

- Two benchmark MDOF nonlinear systems, simulated via Newmark-Beta.

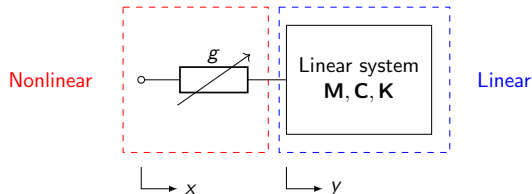


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$$g(u, \dot{u}) = k_1 u + k_3 u^3 + c_1 \dot{u} + c_3 \dot{u}^3$$

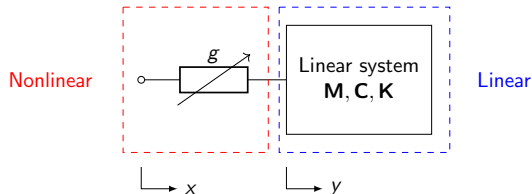


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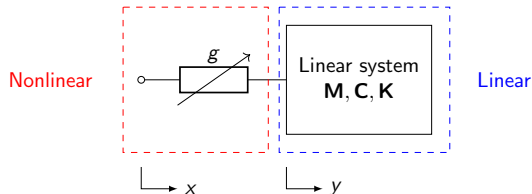


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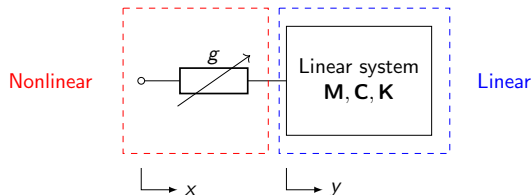


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- $N = 2500$ 'frames' of length $L = 2000$ samples – 80:20 train/test split. NMSE Loss.

Comparison - test loss



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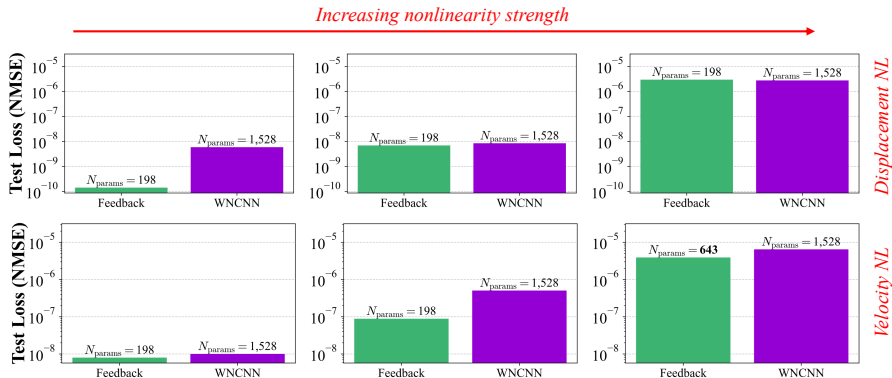


Figure 10: Final test loss comparison between Feedback and WNCNN models.

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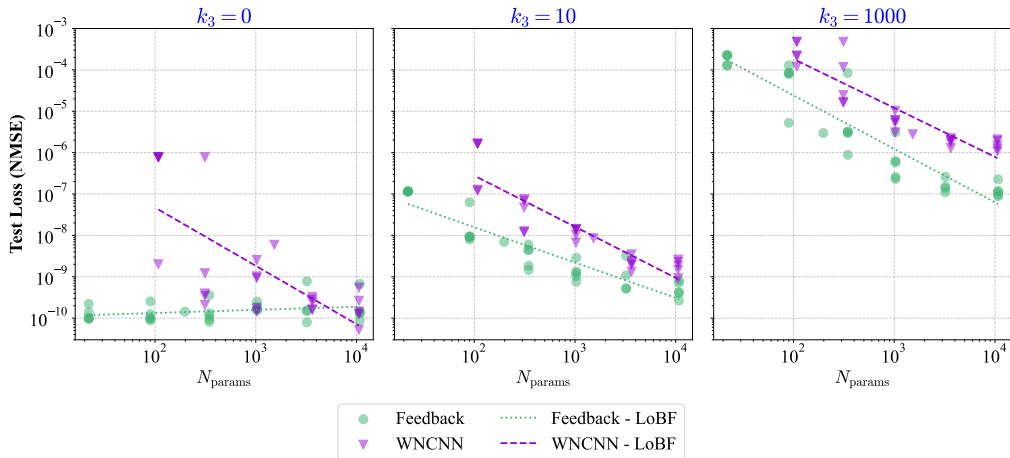


Figure 11: Loss vs number of parameters for the displacement nonlinearity.

Scaling behaviour II

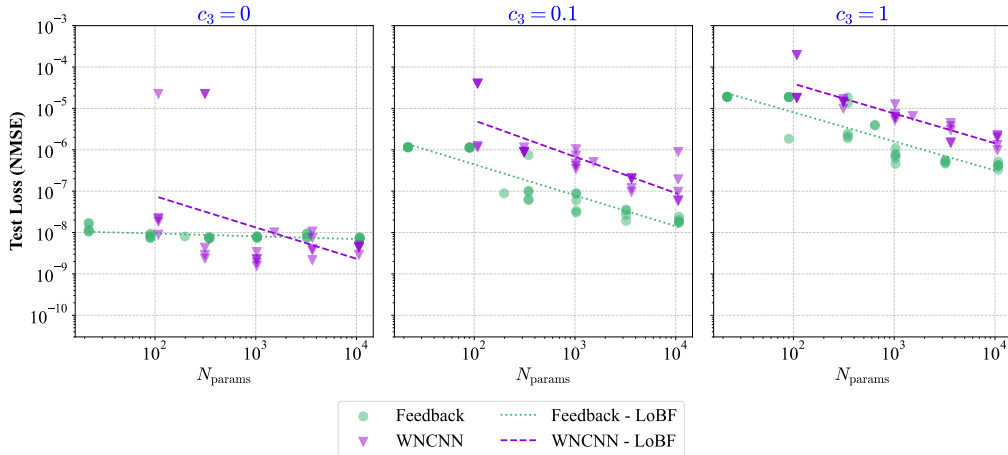


Figure 12: Loss vs number of parameters for the velocity nonlinearity.

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- 1 Feedback models match feedforward model performance with fewer parameters.

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- 1 **Feedback** models match **feedforward** model performance with fewer parameters.
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Conclusions

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Feedback approach shows promise for ANC modelling – both in **speed** and **accuracy**

- Future work on more complex systems:

Conclusions

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- Future work on more complex systems:
 - Plate experimental rig – substructure coupling, MIMO, real noise.

Conclusions

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Feedback approach shows promise for ANC modelling – both in **speed** and **accuracy**

- Future work on more complex systems:
 - Plate experimental rig – substructure coupling, MIMO, real noise.
 - Full scale car – further complexity, practical challenges.

Hybrid physics-ML models for automotive active noise control

ISMA2026, AVC2 – Leuven, Belgium

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¹Cambridge University Engineering Department

²Bose Corporation

Monday 7th September 2026



UNIVERSITY OF
CAMBRIDGE

Department of Engineering

BOSE[®]

Input coherence

$$\text{Input coherence} = \gamma_{xy}^2(\omega) = \frac{|S_{xy}(\omega)|^2}{S_{xx}(\omega)S_{yy}(\omega)} \quad (7)$$

- Describes 'how much of the behaviour can be described by a linear model from x to y '.
- Directly related to the achievable noise reduction in dB: $10 \log(1 - \gamma^2)$.

Output coherence

$$\text{Output coherence} = \gamma_{y_t y_p}^2(\omega) = \frac{|S_{y_t y_p}(\omega)|^2}{S_{y_t y_t}(\omega)S_{y_p y_p}(\omega)} \quad (8)$$

- Describes 'how much of the **nonlinearity is captured** by the model'.

Training and prediction modes

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- A feedback model necessarily operates from time-step to time-step.
- But autoregressive models are often expensive to train, stability issues.
- Therefore, two modes are used:

Training – one step ahead

- Use true output y_t to calculate u for the whole time series:

$$y_p = \mathcal{M}(x, y_t)$$

- Parallel, stable, much faster to train.

Prediction – autoregressive

- Use the model output y_p to calculate u for the next time step:

$$y_p[k] = \mathcal{M}(x[0 : k], y_p[0 : k - 1])$$

- Sequential, realistic for operation.

Example results

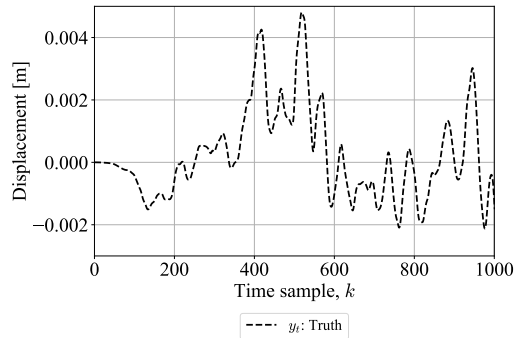
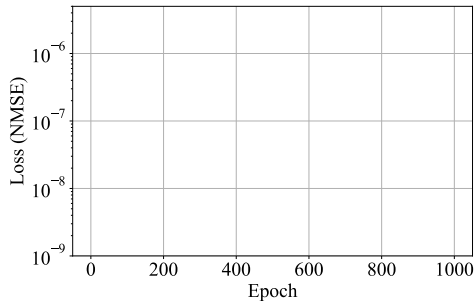


Figure 13: Loss (left) and time series prediction (right) - weak displacement nonlinearity.

Example results

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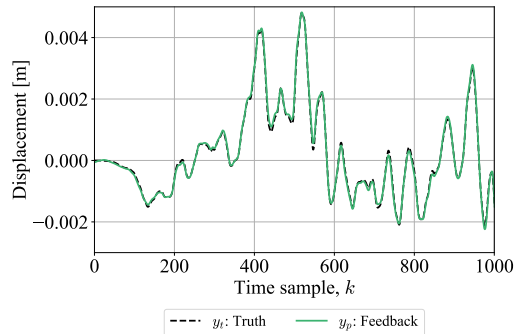
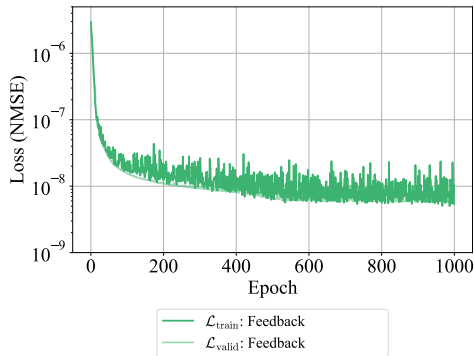


Figure 13: Loss (left) and time series prediction (right) - weak displacement nonlinearity.

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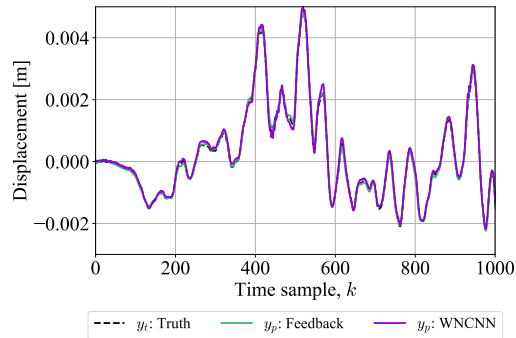
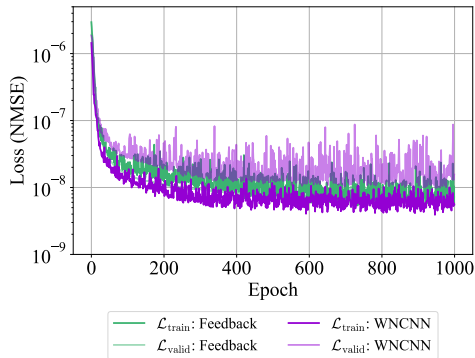


Figure 13: Loss (left) and time series prediction (right) - weak displacement nonlinearity.

Comparison - coherence

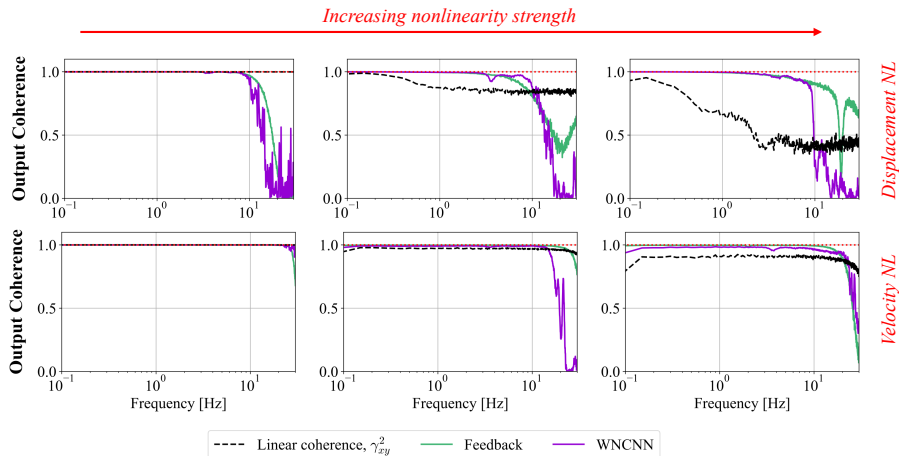


Figure 14: Coherence comparison between **Feedback** and **WNCNN** models.

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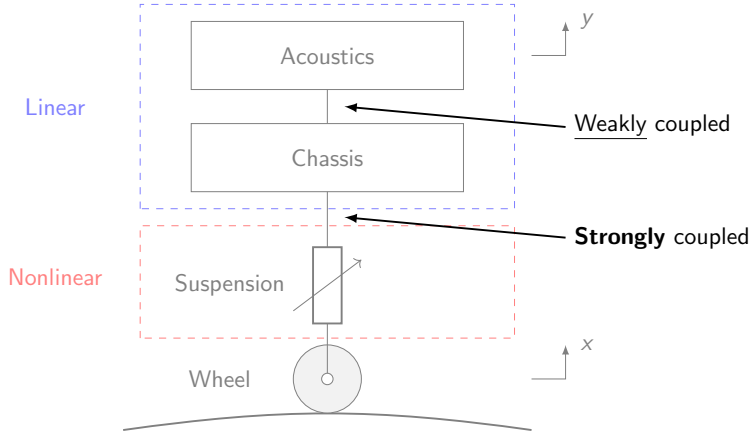


Figure 15: A simplified quarter-car model to illustrate coupling strength.

Additional real world challenges II

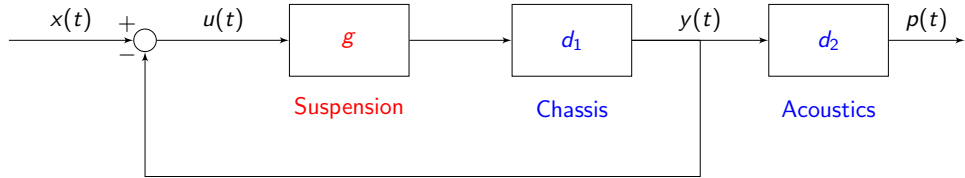


Figure 16: Representation of an automotive system with a weakly coupled acoustic volume.

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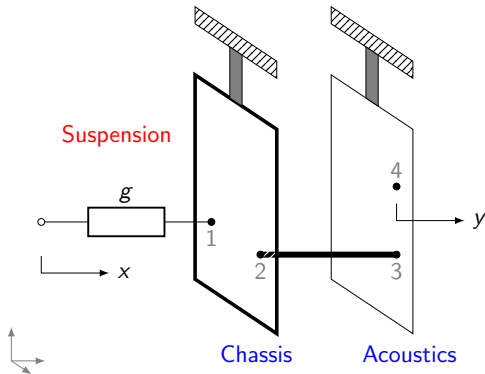


Figure 17: Simplified rig diagram.



Figure 18: Assembled rig.